An Electroencephalography-Based Brain-Computer Interface for Real-Time Control Using Motor Imagery and Deep Learning.

Fernando Rojas
Dept. of Computer Science and Information Engineering
National Taiwan University of Science and Technology
Taipei, Taiwan
rojasmfer@proton.me

Yuan-Hsiang Lin
Dept. of Electronic and Computer Engineering
National Taiwan University of Science and Technology
Taipei, Taiwan
linyh@mail.ntust.edu.tw

***Abstract*—Systems for real-time interactive control using electroencephalography (EEG) have been widely explored for medical, educational, and entertainment purposes. This paper presents the development of a brain-computer interface (BCI) that translates a user's mental commands, detected through EEG signals, into in-game actions, specifically within a Unity environment. The core structure is composed of an EEG acquisition device, a signal processing pipeline for noise reduction and feature extraction, a machine learning framework for classifying brainwaves patterns, and a game interface for command translation. The primary technical challenges identified are the inherent latency of real-time EEG processing and the significance of the model's accuracy for predicting the action to be performed depending on the EEG readings. The proposed implementation involves the use of an accessible EEG device (BioAmp EXG Pill), employing a deep learning model and implementing a communication protocol for integration with a video-game. The project's development roadmap is divided into two phases: an initial phase focused on signal acquisition, model development, and initial third-party videogames implementation, followed by a second phase for system integration, game development, user testing, and refinement. This report covers the first phase of this project.**

***Index Terms*—electroencephalography, brain-computer interface, brainwaves**

I. Introduction

The field of BCIs stands at the forefront of human-technology interaction, offering the potential to augment or alternate from traditional input methods to direct neural commands. This project seeks to contribute and serve as an experiment in this field by developing an EEG-based BCI for video game control. Video games, with their dynamic and interactive nature, serve as an ideal platform for evaluating the real-world performance of a non-invasive BCI [2]. Throughout this report we will present the project's technical framework, how key challenges were addressed, the architecture of the signal processing as well as the deep learning model design, and finally, the results and future work planned for the total completion of the project. The primary motivation is to demonstrate that low-cost, accessible hardware—like the BioAmp EXG Pill and Arduino—can be paired with Deep Learning models to achieve real-time, hands-free interaction with software environments.

II. System Architecture

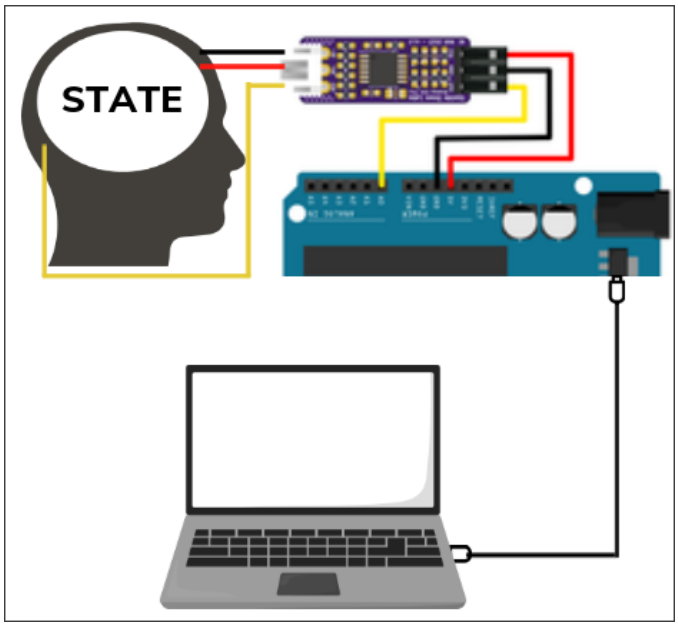

Fig. 1. System overview: EEG signal acquisition from the user via BioAmp EXG Pill, processed by Arduino R3, and transmitted to a laptop for real-time classification.

The system's architecture is a multi-component pipeline designed to process neural signals and translate them into actionable game commands, as illustrated in Fig. 1. The workflow begins with an EEG acquisition device—in this case the BioAmp EXG Pill from UpsideDownLabs [5]—capturing raw electrical signals from the user's scalp via pre-frontal cortex electrodes. These signals are transmitted through an Arduino R3 to a host laptop. Once received, the signals are preprocessed to remove noise and artifacts, and then features can be extracted from the cleaned data. A trained machine learning model classifies these features into specific

mental commands, which are subsequently transmitted to the game interface as key strokes. The architecture is planned to be modular, allowing for independent development and optimization of each component.

A. Electroencephalography Hardware, Signal Quality and Data Acquisition

The project uses the BioAmp EXG Pill (shown in Fig. 2), a product developed by UpsideDownLabs [5] which is an accessible EEG device that suffices the requirements for this case proof-of-concept. While this device has limitations in spatial resolution compared to high-end research systems, its 250 Hz sampling rate is adequate for capturing the key frequency bands of interest. To ensure signal quality despite

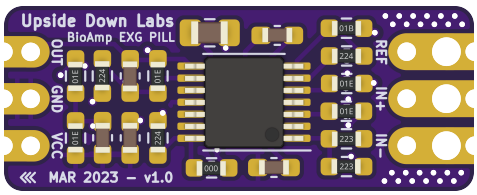

Fig. 2. The BioAmp EXG Pill used for analog-to-digital signal acquisition.

potential noise, experimental protocols were designed to minimize artifacts and the preprocessing pipeline includes the following.

- **Bandpass Filtering**: to remove DC drift and high-frequency noise.
- **Notch Filtering**: to eliminate electrical grid interference depending on the region (e.g., 50/60 Hz).
- **Zero Phase Shift**: Filters are applied forward and backward to ensure no phase distortion in the signal.
- **Segmentation**: Raw data is divided into 1-second windows (250 samples) for model input.

Signal acquisition is achieved by pairing the device with an Arduino R3 and establishing a serial communication via USB port at a baud rate of 115,200 for the data transmission. Then, a Python script records continuous EEG readings for a specified duration (5 minutes) and saves them with timestamps into a CSV file. Data augmentation was implemented to improve generalization and deal with overfitting, the two techniques being jittering and scaling were applied.

B. Environment Setup and User Preparation

The reliability of a BCI system depends heavily on the ”signal-to-noise” ratio established before data processing begins. This section details the physical protocols used to ensure high-quality neural recording.

1) Skin Preparation and Electrode Application: To minimize contact impedance—the resistance between the skin and the electrode—a rigorous skin preparation protocol is followed. The targeted areas of the scalp and forehead are first cleaned using 70 percent isopropyl alcohol to remove skin oils, dead dermis cells, and sweat, which act as insulators. Following cleaning, a conductive EEG gel is applied to the electrodes. This gel serves as the transducer medium, facilitating the flow of ions from the skin to the BioAmp EXG Pill’s sensors.

2) The International 10 20 System: Electrode placement follows the International 10-20 System, a standardized method used to ensure that electrode locations are proportional to the skull’s size and shape. According to Homan et al. (1987) [1], the 10-20 system provides a reliable correlation between scalp coordinates and underlying cerebral structures. Specifically, Homan’s research using CT scans demonstrated that while there is some variability, the system accurately targets specific cortical regions across different subjects.

For this project, the Frontal-polar (Fp) position is used. This location is chosen for two primary reasons:

- Relevance to mental state: The prefrontal cortex is heavily involved in attention related functions, including sustained concentration (focused state) versus mediation or relaxation (relaxed state).
- Accessibility: The forehead is a hairless region, allowing for consistent contact between electrodes and skin without the interference of hair follicles.

3) Electromagnetic Interference (EMI) Mitigation: EEG signals are measured in microvolts, making them extremely susceptible to environmental noise. To achieve the ”cleanest” possible raw signal before software filtering, the following is applied:

- Whenever possible, the computer used for data acquisition should be disconnected from any AC power supply and run solely on battery power.
- The subject and the acquisition hardware should ideally be positioned at least 5 meters away from active power sockets or large transformers.

III. Machine Learning Framework

The BCI system utilizes a 1D Convolutional Neural Network (CNN) designed to extract temporal features from raw EEG sequences. Unlike traditional machine learning approaches that require manual feature engineering, the 1D CNN automatically learns relevant spatial-temporal patterns directly from the voltage-time series.

A. Architectural Specification

The project employs a 1D CNN optimized for binary classification, using the Keras API [6] within the TensorFlow framework [7] as the primary environment for model development. The input layer accepts a tensor of shape $(250, 1)$, representing a 1-second epoch sampled at 250 Hz.

- **Feature Extraction Layers**: Two consecutive convolutional blocks are employed. Each block consists of a 1D Convolutional layer with a kernel size of 5 to capture local temporal dependencies. Batch Normalization is applied after each convolution to accelerate convergence and mitigate internal covariate shift.
- **Dimensionality Reduction**: Max-Pooling layers with a pool size of 4 follow each convolutional block, reducing the feature map size while maintaining the most salient activations.

- **Regularization**: To prevent overfitting given the high variance of EEG data, a progressive dropout strategy is used: 20% in the convolutional blocks and 50% in the final dense layer.
- **Classification Head**: The flattened feature vector is fed into a 50-unit fully connected (Dense) layer with L_2 regularization ($\lambda = 0.01$). The final output is a single neuron with a Sigmoid activation function, producing a probability score $y \in [0, 1]$ for binary classification.

IV. Implementation and Results

During the development, the model underwent several iterations to balance accuracy and overfitting. Using this architecture and data augmentation, the model achieved a training accuracy of 73.7% and a validation accuracy of 72.8%, with significantly improved loss stability (val_loss: 0.59). To test the model performance in a real scenario we implemented live inferencing combined with a third party game integration (specifically Evo-F4 [3]) by using the model's predictions of the state and translating it into keyboard commands using ydotool [4]. The "focused" state triggered the movement of the player and the "relaxed" state simulated the brakes.

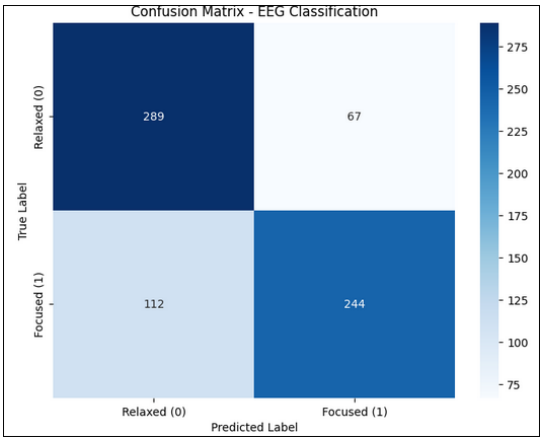

Fig. 3. Confusion Matrix for binary classification of Focused vs. Relaxed states.

The performance of the 1D CNN is further elucidated by the confusion matrix (Fig. 3). Analysis of the confusion matrix reveals a bias toward the "Relaxed" state (Recall: 0.81), indicating that the system is highly effective at identifying the 'Stop' command. However, the Focused state (Recall: 0.69) shows a higher rate of False Negative. These results achieve the acceptable level of accuracy with some level of error which is common in EEG data due to the non-stationary nature of the signals and potential muscle artifact contamination in the dataset.

V. Conclusion and Future Work

Developing an EEG-controlled interactive entertainment system resulted to be a challenging yet achievable capstone project. By adopting a methodical approach to hardware, software, and algorithm design, the work successfully demonstrates the feasibility of a non-invasive, accessible BCI for entertainment applications. The next phase of development will prioritize system integration and game development, establishing a communication pipeline between the BCI backend and the Unity Engine.

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